

23

定積分の置換積分法

$\alpha < \beta$ のとき、区間 $[\alpha, \beta]$ で微分可能な関数 $x = g(t)$ に対し

$a = g(\alpha)$ 、 $b = g(\beta)$ ならば

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(g(t)) g'(t) dt$$

※ 理屈はあとで

(例1) 次の定積分を求めよ。

(1) $\int_0^1 x(2x+1)^3 dx$

$2x+1 = t$ とおくと、 $x = \frac{1}{2}(t-1)$ より

$$dx = \frac{1}{2} dt$$

$$\begin{array}{c|c} x & 0 \rightarrow 1 \\ \hline t & 1 \rightarrow 3 \end{array}$$

であるから

$$\begin{aligned} (\text{与式}) &= \int_1^3 \frac{1}{2}(t-1)t^3 \cdot \frac{1}{2} dt \\ &= \frac{1}{4} \int_1^3 (t^4 - t^3) dt \\ &= \frac{1}{4} \left[\frac{1}{5}t^5 - \frac{1}{4}t^4 \right]_1^3 \\ &= \frac{1}{4} \left\{ \frac{1}{5}(243-1) - \frac{1}{4}(81-1) \right\} \\ &= \frac{71}{16} \text{ 〃} \end{aligned}$$

(2) $\int_{-1}^2 \frac{x}{\sqrt{3-x}} dx$

$\sqrt{3-x} = t$ とおくと、 $x = 3-t^2$ より

$$dx = -2t dt$$

$$\begin{array}{c|c} x & -1 \rightarrow 2 \\ \hline t & 2 \rightarrow 1 \end{array}$$

であるから

$$\begin{aligned} (\text{与式}) &= \int_2^1 \frac{3-t^2}{t} \cdot (-2t) dt \\ &= 2 \int_1^2 (3-t^2) dt \\ &= 2 \left[3t - \frac{1}{3}t^3 \right]_1^2 \\ &= 2 \left\{ 3(2-1) - \frac{1}{3}(8-1) \right\} \\ &= \frac{4}{3} \text{ 〃} \end{aligned}$$

(例2) 次の定積分を求めよ。

(1) $\int_1^5 \frac{1}{x\sqrt{x+4}} dx$

$\sqrt{x+4} = t$ とおくと、 $x = t^2 - 4$ より

$$dx = 2t dt$$

$$\begin{array}{c|c} x & 1 \rightarrow 5 \\ \hline t & \sqrt{5} \rightarrow 3 \end{array}$$

であるから

$$\begin{aligned} (\text{与式}) &= \int_{\sqrt{5}}^3 \frac{1}{(t^2-4)t} \cdot 2t dt \\ &= 2 \int_{\sqrt{5}}^3 \frac{1}{t^2-4} dt \\ &= 2 \cdot \frac{1}{4} \int_{\sqrt{5}}^3 \left(\frac{1}{t-2} - \frac{1}{t+2} \right) dt \\ &= \frac{1}{2} \left[\log \left| \frac{t-2}{t+2} \right| \right]_{\sqrt{5}}^3 \\ &= \frac{1}{2} \left(\log \frac{1}{5} - \log \frac{\sqrt{5}-2}{\sqrt{5}+2} \right) \\ &= \frac{1}{2} \log \frac{\sqrt{5}+2}{5(\sqrt{5}-2)} \\ &= \frac{1}{2} \log \frac{(\sqrt{5}+2)^2}{5} \\ &= \log \frac{5+2\sqrt{5}}{5} \text{ 〃} \end{aligned}$$

(2) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{\sin x} dx = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sin x}{\sin^2 x} dx$
 $= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sin x}{1-\cos^2 x} dx$

そこで、 $\cos x = t$ とおくと

$$-\sin x dx = dt \quad \text{つまり} \quad \sin x dx = -dt$$

$$\begin{array}{c|c} x & \frac{\pi}{3} \rightarrow \frac{\pi}{2} \\ \hline t & \frac{1}{2} \rightarrow 0 \end{array}$$

であるから

$$\begin{aligned} (\text{与式}) &= \int_{\frac{1}{2}}^0 \frac{1}{1-t^2} \cdot (-dt) \\ &= \int_0^{\frac{1}{2}} \frac{1}{1-t^2} dt \\ &= \frac{1}{2} \int_0^{\frac{1}{2}} \left(\frac{1}{1-t} + \frac{1}{1+t} \right) dt \\ &= \frac{1}{2} \left[\log \left| \frac{1+t}{1-t} \right| \right]_0^{\frac{1}{2}} \\ &= \frac{1}{2} \log 3 \text{ 〃} \end{aligned}$$

(3) $\int_0^1 \frac{1}{e^x + 2e^{-x} + 3} dx$

$e^x = t$ とおくと、 $x = \log t$ より

$$dx = \frac{1}{t} dt$$

$$\begin{array}{c|c} x & 0 \rightarrow 1 \\ \hline t & 1 \rightarrow e \end{array}$$

であるから

$$\begin{aligned} (\text{与式}) &= \int_1^e \frac{1}{t + \frac{1}{t} + 3} \cdot \frac{1}{t} dt \\ &= \int_1^e \frac{1}{(t+1)(t+2)} dt \\ &= \int_1^e \left(\frac{1}{t+1} - \frac{1}{t+2} \right) dt \\ &= \left[\log \left| \frac{t+1}{t+2} \right| \right]_1^e \\ &= \log \frac{e+1}{e+2} - \log \frac{2}{3} \\ &= \log \frac{3(e+1)}{2(e+2)} \text{ 〃} \end{aligned}$$